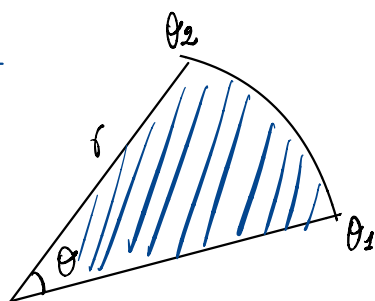


Lecture 2, Mar 18 2026.

10.4 Calculus in Polar Coordinates.

Def. A function in polar coordinates is a function $r = f(\theta)$.

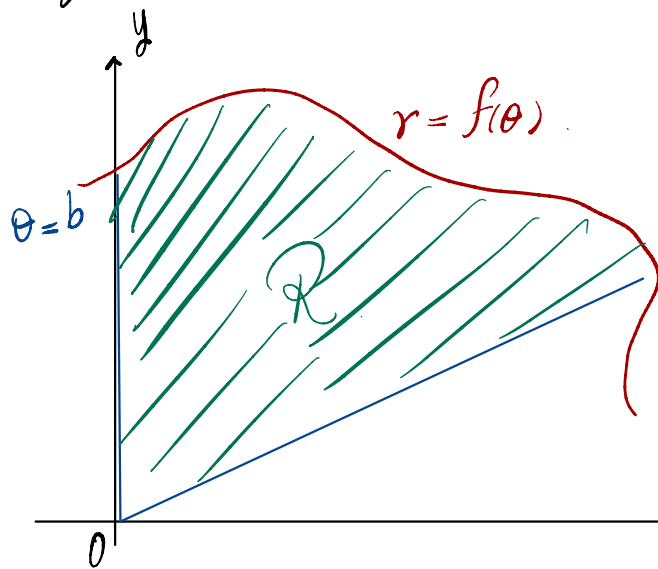
Area.



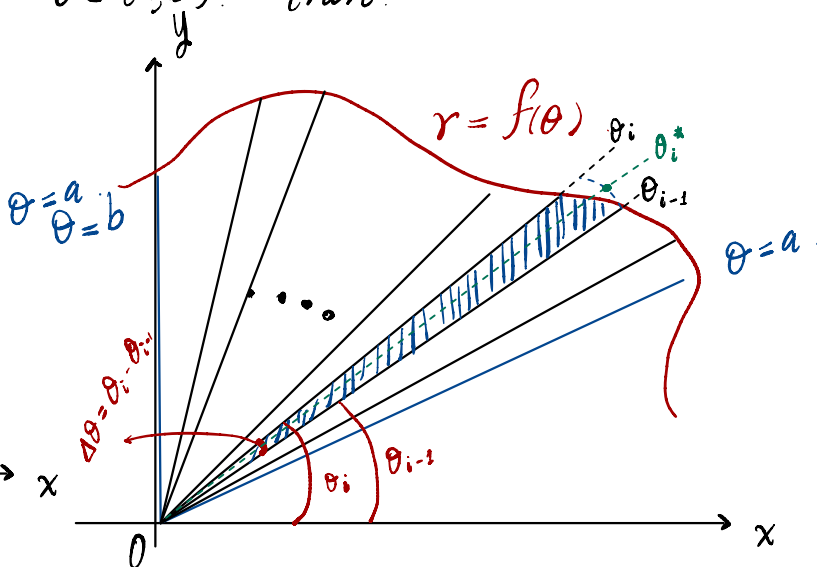
then the area of region is

$$A = \frac{1}{2} r^2 \theta.$$

In general.



Let consider the region R and $\theta \in [a, b]$ then.



then $\Delta A_i \approx \frac{1}{2} [r]^2 \Delta \theta$. but $r = f(\theta_i^*)$. then $\Delta A_i \approx \frac{1}{2} [f(\theta_i^*)]^2 \Delta \theta_i$
then

$$A \approx \sum_{i=1}^n \frac{1}{2} [f(\theta_i^*)]^2 \Delta \theta_i.$$

Def. Let $r = f(\theta)$ be a continuous function and $\theta \in [a, b]$ then the area "under the curve" is defined in Riemann sense as.

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{2} [f(\theta_i^*)]^2 \Delta \theta_i = \int_a^b \frac{1}{2} [f(\theta)]^2 d\theta.$$

Example. Find the area enclosed by one loop of the four-leaved rose

$$r = \cos 2\theta$$

$$r = \cos 2\theta$$

In this case $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$ then

$$A = \int_{-\pi/4}^{\pi/4} \frac{1}{2} (\cos 2\theta)^2 d\theta$$

$$A = 2 \left(\frac{1}{2} \right) \int_0^{\pi/4} \cos^2 \theta d\theta$$

$$A = \frac{1}{2} \int_0^{\pi/4} (1 + \cos 4\theta) d\theta = \frac{1}{2} \int_0^{\pi/4} d\theta + \frac{1}{2} \int_0^{\pi/4} \cos 4\theta d\theta = \frac{\pi}{8} + \frac{1}{2} \int_0^{\pi/4} \cos 4\theta d\theta$$

$$A = \frac{\pi}{8} + \frac{1}{8} \sin 4\theta \Big|_0^{\pi/4} = \frac{\pi}{8}$$

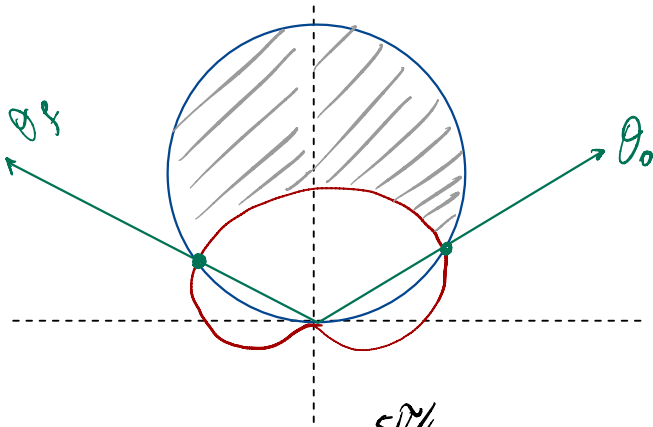
Example. Find the area of the region that lies inside the circle

$r = 3 \sin \theta$ and outside the cardioid $r = 1 + \sin \theta$.

$$\text{Solving } 1 + \sin \theta = 3 \sin \theta$$

$$\text{then } \sin \theta = \frac{1}{2}$$

$$\text{then } \theta_0 = \frac{\pi}{6} \text{ and } \theta_f = \frac{5\pi}{6}$$



$$A = \frac{1}{2} \int_{\pi/6}^{5\pi/6} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\pi/6}^{5\pi/6} (1 + \sin \theta)^2 d\theta$$

$$A = \frac{9}{2} \int_{\pi/6}^{5\pi/6} \sin^2 \theta d\theta - \frac{1}{2} \int_{\pi/6}^{5\pi/6} (1 + 2 \sin \theta + \sin^2 \theta) d\theta$$

$$A = \left(\frac{9}{2} - \frac{1}{2} \right) \int_{\pi/6}^{5\pi/6} \sin^2 \theta d\theta - \frac{1}{2} \int_{\pi/6}^{5\pi/6} d\theta - \int_{\pi/6}^{5\pi/6} \sin \theta d\theta$$

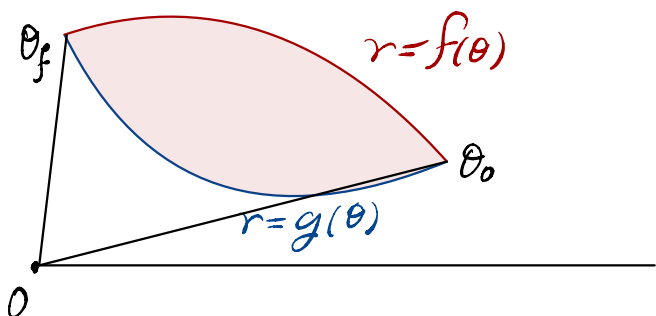
$$A = 4 \int_{\pi/6}^{5\pi/6} \frac{1}{2} (1 - \cos 2\theta) d\theta - \frac{1}{2} \left(\frac{5\pi}{6} - \frac{\pi}{6} \right) + \cos \theta \Big|_{\pi/6}^{5\pi/6}$$

$$A = 2 \left(\frac{5\pi}{6} - \frac{\pi}{6} \right) - \sin 2\theta \Big|_{\pi/6}^{5\pi/6} - \frac{\pi}{3} - \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}$$

$$A = 4\frac{\pi}{3} - \underbrace{\left(\sin \frac{5\pi}{3} - \sin \frac{\pi}{3} \right)}_0 - \frac{\pi}{3} - \sqrt{3}$$

$$A = \pi - \sqrt{3}$$

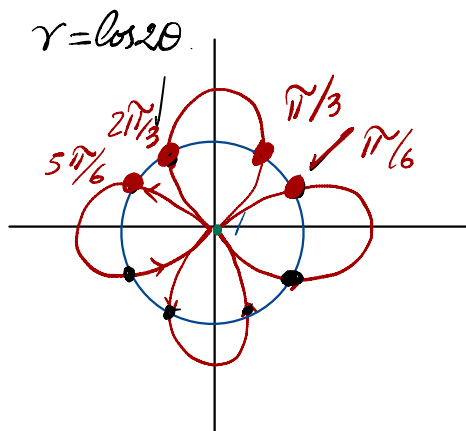
In general the one between polar curves.



$$A = \frac{1}{2} \int_{\theta_0}^{\theta_f} [f^2(\theta) - g^2(\theta)] d\theta.$$

Example. Find all points of intersection of the curves $r = \cos 2\theta$ and $r = \frac{1}{2}$.

Sln. Solving $\cos 2\theta = \frac{1}{2}$. then $2\theta = \frac{\pi}{3} \Rightarrow \theta = \frac{\pi}{6}$
 $\left(\frac{1}{2}, \frac{\pi}{6} \right), \left(\frac{1}{2}, -\frac{\pi}{6} \right)$



$$2\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

then $\left(\frac{1}{2}, \frac{5\pi}{6} \right), \left(\frac{1}{2}, \frac{7\pi}{6} \right), \left(\frac{1}{2}, \frac{11\pi}{6} \right)$
 $\left(\frac{1}{2}, \frac{\pi}{6} \right), \left(\frac{1}{2}, \frac{2\pi}{3} \right), \left(\frac{1}{2}, \frac{4\pi}{3} \right)$